A-A-53608
10 August 1988

COMMERCIAL ITEM DESCRIPTION

HYDROMETERS, GRADUATED SCALE
(With Thermometer, API Gravity)

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-H-36343B which will be cancelled.

This commercial item description covers glass, graduated scale hydrometers, with integral thermometers, suitable for determining the American Petroleum Institute (API) gravity of petroleum products and other liquids having a surface tension not exceeding 33 dynes per centimeter.

Salient characteristics.

The hydrometers shall be of the following types, classes, and styles:

Type I. API Gravity Thermo-Hydrometers; Thermometer scale in body.

The API gravity range in degrees shall be designated by an ASTM Hydrometer Number per ASTM E100 as follows:

ASTM Hydrometer Number	Nominal API Gravity Range in Degrees
51 H	-1 to +11
52 H	9 to 21
53 H	19 to 31
54 H	29 to 41
55 H	39 to 51
56 H	49 to 61
57 H	59 to 71
58 H	69 to 81
59 H	79 to 91
60 H	89 to 101

AMSC N/A FSC 6630

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Classification by the thermometer range in degrees Fahrenheit (°F) shall be designated by a suffix to the ASTM Hydrometer Number per ASTM E100 as follows:

Class 1. ASTM Suffix L Thermometer range zero to 150°F.
Class 2. ASTM Suffix M Thermometer range 30 to 180°F.
Class 3. ASTM Suffix H Thermometer range 60 to 220°F.

Type II. API Gravity Thermo-Hydrometers; Thermometer scale in body.

The hydrometer design shall comply to Figure 1b of ASTM E100.

Classification by thermometer range in degrees Fahrenheit (°F) shall be as follows:

Class 1. Thermometer range zero to 110°F.
Class 2. Thermometer range 20 to 130°F.

The API gravity range in degrees shall be designated by style number as follows:

Style	Nominal API Gravity Range in Degrees
1	-1 to +11
2	9 to 21
3	19 to 31
4	29 to 41
5	39 to 51
6	49 to 61
7	59 to 71
8	69 to 81

The materials used in the construction of Type I and Type II hydrometers shall be in accordance with ASTM E100.

Type I hydrometers shall be constructed with dimensions, thermometers, temperature scales and API gravity scales in accordance with ASTM E100. The error in Fahrenheit (F.) temperature scale shall not be greater than 1°F when tested as specified in ASTM E126. The error in API gravity scale shall not be greater than 0.1° API when tested as specified in ASTM E126.

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Type II hydrometers shall consist of a glass stem and glass plummet, with an integral thermometer. The plummet end shall be filled with ballast as specified in ASTM E100. The thermometer scale shall be located in the body. The scale, thermometer, and ballast shall be held securely in place within the hydrometer. The thermometer bulb shall be embedded in a suitable heat transfer medium enclosed within the body of the hydrometer.

The overall length of Type II hydrometers shall be 190 ± 5 millimeters (mm). The bulb diameter shall be 16.5 ± 2 mm with a stem size of 4.25 mm.

Thermometers of Type II hydrometers shall be graduated in 2°F increments with major interval lines at 10°F and with every 20°F interval numbered.

The error in Fahrenheit (F) temperature scale for Type II hydrometers shall not be greater than 2°F when tested in accordance with ASTM E126.

The API gravity scale for Type II hydrometers shall be in 0.2° API increments, with major interval lines and numbers every 1° API.

The error in API gravity scale for Type II hydrometers shall be not greater than 0.2° API when tested in accordance with ASTM E126.

The following information shall be legibly and permanently marked in a readily visible location on the inside of the hydrometer: Units of hydrometer scale, standard temperature (60°F), ASTM Hydrometer Number for Type I, and manufacturer's serial number and name or registered trademark.

Workmanship. The hydrometers shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One hydrometer of the type, class, and style specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities

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suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspections and tests. Inspections and tests shall be conducted in accordance with commercial practice to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for inspections and tests. Sampling for inspections and tests shall be conducted in accordance with commercial practice. The unit of product for test purposes shall be one hydrometer of specified type, class and style.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/ contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal

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Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	API Range	Type	Class	ASTM No.
6630-00-242-9257	19 to 31°	I	2	53H
6630-00-242-9258	29 to 41°	I	2	54H
6630-00-245-8374	59 to 71°	I	1	57H
6630-00-245-8375	69 to 81°	I	1	58H
6630-00-245-8376	39 to 51°	I	1	55H
6630-00-245-8377	49 to 61°	I	1	56H
6630-00-245-8378	-1 to 11°	I	3	51H
6630-00-245-8379	9 to 21°	I	3	52H
6630-00-252-2264	79 to 91°	I	1	59H

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NSN	API Range	Type	Class	Style
6630-00-265-7610	9 to 21°	II	2	2
6630-00-265-7611	19 to 31°	II	2	3
6630-00-265-7758	29 to 41°	II	2	4
6630-00-265-7759	39 to 51°	II	2	5
6630-00-265-7764	49 to 61°	II	1	6
6630-00-265-7765	59 to 71°	II	1	7
6630-00-815-2267	69 to 81°	II	1	8
6630-00-815-2270	-1 to 11°	II	2	1

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.)

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6630-0475
Location: Disk # 6

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53608	HYDROMETERS, GRADUATED SCALE (With Thermometer, API Gravity)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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